

A.W. YESHARA AMUNUGAMA

BSc (Hons) Computer Science Student

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Location: Kadawatha, Gampaha, Sri Lanka

PROFESSIONAL SUMMARY

Final-year BSc (Hons) Computer Science student at the University of Bedfordshire, with a Higher National Diploma in Software Engineering and a Diploma in Computer System Designing from NIBM. Experienced in full-stack development, mobile applications, AI-integrated systems, and Agile project management. Developed and delivered multiple end-to-end projects using React Native, Node.js, Python, Java, Firebase, Supabase, and machine learning tools. Seeking a junior software development, QA, or graduate technology role where I can contribute to impactful products.

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, C#, C++, SQL, R, Dart, HTML, CSS/SCSS
Frameworks & Libraries: React Native, React, Next.js, Node.js, Express.js, FastAPI, Flutter, Vite, Tailwind CSS
Databases & Cloud: Firebase, Firestore, Supabase, PostgreSQL, MySQL, SQLite
AI / Data Tools: FAISS, TinyLlama/LLaMA, LoRA, Ollama, RAG, Machine Learning, R Language
Tools & Platforms: GitHub, Postman, ClickUp, Figma, Expo Go, Arduino IDE, IntelliJ IDEA, NetBeans
Testing: Functional Testing, Integration Testing, UAT, API Testing, Regression Testing
Methodologies: Agile Scrum, PRINCE2 Agile, OOP, MVC, DAO Pattern, REST APIs, UML Diagrams

EDUCATION

BSc (Hons) Computer Science	Undergraduate (Expected 2026)
University of Bedfordshire (UK) – in partnership with SLIIT City University, Sri Lanka	
Higher National Diploma in Software Engineering	2024– 2025
National Institute of Business Management (NIBM) GPA: 3.5 / 4.0	
Diploma in Computer System Design	2022 – 2023
National Institute of Business Management (NIBM) GPA: 3.85 / 4.0 (Distinction)	
Certificate Course in Computer Science	2022 – 2023
National Institute of Business Management (NIBM) Grade: A Pass	
A-Level – Mathematics Stream	2018 – 2022
Viharamahadevi Balika Vidyalaya, Kiribathgoda Completed 2022 (2021 A/L Batch)	

PROJECTS

FlavorMind – AI-Powered Culinary Assistant Mobile Application	2025 – 2026
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Final Year Thesis Project | BSc (Hons) Computer Science | University of Bedfordshire

- Developed a full-stack AI-powered mobile culinary assistant using React Native Expo, TypeScript, Node.js Express backend, and Python FastAPI AI service to help users recreate recipes from remembered taste, smell, texture, or ingredient descriptions.
- Implemented memory-based recipe recall using FAISS vector search and retrieval-augmented generation (RAG) to semantically match food memory descriptions with similar recipes, achieving 95% semantic match accuracy.

- Fine-tuned a TinyLlama 1.1B model using LoRA for structured recipe generation, local ingredient adaptation, and dynamic serving-size scaling tasks.
- Developed a Global-to-Local Ingredient Adaptation Engine to suggest Sri Lankan alternatives for unavailable global ingredients, supporting localised cooking.
- Integrated Firebase Authentication and Cloud Firestore for user profiles, recipe storage, personal cookbooks, community sharing, feedback, and admin moderation workflows.
- Built a web-based admin panel using Vite React, TypeScript, and Tailwind CSS for approving and managing recipes, cookbooks, and ingredient substitution data.
- Conducted functional testing, AI evaluation, and user acceptance testing; achieved 89 passed test cases and positive UAT feedback from beta participants.

Tech Stack: React Native, Expo, TypeScript, Node.js, Express.js, Python, FastAPI, Firebase, Firestore, FAISS, TinyLlama/LLaMA, LoRA, Ollama, REST APIs, GitHub, Expo Go

SpellBeat / Word Hearts Challenge – JavaFX Desktop Word Game

2025

DSA Project | Java Desktop Application

- Developed a full-featured JavaFX desktop word game combining vocabulary-based word guessing with a heart mini-game energy system, supporting 12 progressive difficulty levels.
- Implemented secure user authentication with salted SHA-256 password hashing, login rate limiting, account lockout, forgot password flow, and Remember Me functionality.
- Integrated multiple external APIs including a Random Word API, Wordnik for definitions and examples, Datamuse for synonyms, and a Marc Conrad heart puzzle API for mini-game content.
- Designed and implemented an MVC-like architecture with separate controller, service, database, model, UI, and utility packages for clean separation of concerns.
- Built persistent game state management using SQLite with JDBC, storing user profiles, level progression, heart counts, and game history.

Tech Stack: Java 21, JavaFX, SQLite, JDBC, JSON, Gson, CSS, External REST APIs

Custom AI Chatbot – Sinofetch Corporation

2025

Agile Group Project | Project Manager

- Led and managed a 10-week Agile Scrum project across four sprints to deliver a full-stack AI chatbot prototype with an embeddable website chat widget, Express.js backend API, Supabase database, OpenAI integration, and admin dashboard.
- Coordinated sprint planning, backlog prioritisation, task allocation, team communication, risk management, and formal project documentation including RACI matrix, lessons learned report, and closure report.
- Managed quality assurance activities across chat bubble, admin panel, and API modules; coordinated 29 planned test cases with 8 defects identified and 75% detected during planned QA/testing cycles.
- Oversaw integration of Supabase Authentication and PostgreSQL for admin login, lead storage, conversation persistence, and message history.

Tech Stack: Node.js, Express.js, TypeScript, React, Next.js, Electron, Vite, Tailwind CSS, Supabase, PostgreSQL, OpenAI API, REST APIs, GitHub, ClickUp

Donation & Volunteer Support Platform – Hope

2024 - 2025

Full-Stack Web Development Project

- Developed a full-stack web and admin platform to connect donors with individuals in need including patients, children, and elders.
- Implemented key features including profile approval workflows, donation tracking, volunteer registration, event planning, and income analytics via service fee reporting.

Food Ordering App – Flavor Fiesta

Sep 2024 – Oct 2024

Mobile Application

- Built a mobile application for managing food orders with user and admin roles, real-time order updates, Google Maps location tracking, image uploads, and promotional features.

Tech Stack: Flutter, Firebase, Dart

Gym Management System – ABI Fitness

Aug 2024 – Sep 2024

Web Application

- Developed a web-based system for gym operations including member registration, subscription handling, trainer allocation, and class scheduling.

Tech Stack: React, IntelliJ IDEA

Employee Management System for Construction Company

2023 – 2024

Web Application

- Developed a web-based HR management system including employee registration, attendance tracking, leave management, KPI measurement, task performance monitoring, trainee supervision, safety protocols, promotion workflows, and strategic reporting.

Additional Projects

- Robotic Package Sorting System – Contributed to an automated sorting robot with Bluetooth modules for real-time data sharing (Arduino IDE, Sensors).
- Credit Limit Analysis – Conducted statistical data analysis using regression models in R to determine factors influencing credit limits.
- Flight Reservation System – Built a command-line C++ application for managing flight bookings, registration, and ticket handling.
- Emergency Assistance App for Heart Patients – Created a web app using C# to alert emergency contacts and hospitals with user authentication and health data submission.
- Smart Lecture Hall Lighting System – Developed an automated sensor-based lighting system using Arduino with temperature and motion monitoring.

ADDITIONAL INFORMATION

Languages: Sinhala (Fluent), English (Intermediate), Tamil (Basic)

Soft Skills: Teamwork, Leadership, Effective Communication, Critical Thinking, Time Management

Interests: Software Development, AI-Integrated Applications, QA Testing, Full-Stack Engineering

Available for graduate roles, internships, and entry-level technology positions.

REFERENCES

Mr. Nisal Gunadasa | Lead Software Engineer, IFS | Tel: +94 750 785 064

Ms. H.A.S. Sakunika Karunarathne | Senior Software Engineer & Team Lead, Visionex Digital | Tel: +94 768 423 096